INSTRUCTOR BRIEFING SHEET

Section 1: SIEM Tool

Microsoft Sentinel

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a. Name of Tool

Microsoft Sentinel

b. Website

https://azure.microsoft.com/en-ca/products/microsoft-sentinel

c. Version / License

Free trial via personal Microsoft Azure account. New accounts receive $200 USD in credits for 30 days. Microsoft Sentinel provides an additional 31-day free trial period during which data ingestion charges up to 10 GB/day are waived. Activated April 18, 2026, valid until May 19, 2026. After the trial, pricing is consumption-based (pay-per-GB ingested). No software installation is required as Sentinel is fully cloud native.

d. Main Features of the Tool

- Cloud-native SIEM/SOAR platform hosted on Microsoft Azure which can be accessible from any browser with no on-premises infrastructure required
- Log Analytics Workspace (CYBR3090-Workspace) serves as the central data store where all ingested logs are stored and queried using KQL (Kusto Query Language)
- Microsoft Monitoring Agent (MMA) installed on the Windows 10 sandbox VM forwards Windows event logs and heartbeat signals to the workspace in real time
- Heartbeat monitoring provides continuous visibility into connected devices, the Win10 VM (DESKTOP-MM1IHBE) sent 82 heartbeat signals over 10 hours of uptime during this demonstration
- KQL query editor enables custom threat-hunting queries across all ingested data, used to query device uptime, event counts, and baseline behavior
- Analytics rules engine allows creation of scheduled detection rules that automatically generate incidents when thresholds are exceeded, demonstrated with a custom VM Heartbeat Monitor rule
- Content Hub provides one-click installation of pre-built security solutions including data connectors, analytics rules, and workbooks
- Incident management dashboard correlates related alerts into actionable security incidents with severity scoring (High, Medium, Low)

e. Use in a Production Environment

In a real enterprise environment, Microsoft Sentinel would be deployed as the central security monitoring hub. All endpoints (Windows/Linux workstations, servers), network devices (firewalls, routers, switches), identity providers (Azure Active Directory), and cloud workloads would be connected via data connectors. Security analysts would use the Incidents queue as their daily triage dashboard. Custom KQL hunting queries would run on schedule to proactively search for indicators of compromise. Analytics rules would automate tier-1 detection for example, alerting when a machine stops sending heartbeats (potential outage or compromise), when multiple failed logins occur (brute-force attempt), or when unusual outbound traffic is detected. Playbooks would automate responses such as disabling compromised accounts or blocking malicious IPs. Sentinel's workbooks would generate compliance reports aligned with frameworks such as NIST CSF and ISO 27001.

f. Lessons Learned

- The Log Analytics Workspace must be created before Sentinel can be added — they are separate resources that are linked together during setup
- The Microsoft Monitoring Agent (MMA) is the legacy connection method for non-Azure VMs and remains the simplest option for local Hyper-V machines without Azure Arc
- Sentinel's Content Hub and Data Connectors have moved to the Microsoft Defender portal (security.microsoft.com) — the classic Azure portal redirects automatically
- KQL has a learning curve, but the built-in query templates and IntelliSense make it approachable basic filter, summarize, and project operators are sufficient for meaningful analysis